

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Plastic plug, Part Number 4919495
- Anti-seize compound, Part Number 3824879, or equivalent.

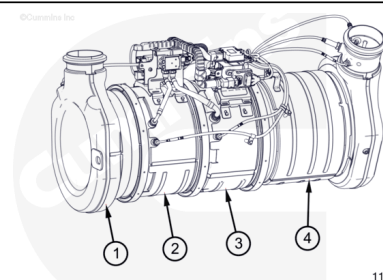
Additional Service Items

- Permanent marker
- Inspection mirror
- Flashlight
- HEPA vacuum
- Vacuum bag (suitable to collect dust and soot)
- Rubber pipe reducer 51 to 102 mm [2 to 4 in].

General Information

⚠ WARNING ⚠

The material captured in a partial flow diesel particulate filter and/or a diesel particulate filter may contain elevated concentrations of metals. Primarily zinc, molybdenum, and possibly polynuclear aromatic hydrocarbons, that may be regulated. These materials must be characterized, handled, and disposed of according to applicable local regulations. In addition, due to the presence of the above-listed chemicals and other potentially toxic components such as oxides of calcium, zinc, phosphorous, silicon, sulfur, and iron, exhaust filter maintenance must be completed only by appropriately trained personnel.



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⚠ CAUTION ⚠

The aftertreatment diesel oxidation catalyst (DOC) elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment

system as damage to the aftertreatment DOC element can result.

⚠ CAUTION ⚠

Do not steam clean the inlet of the DOC as damage to the aftertreatment DOC can result.

Due to the number of exhaust aftertreatment applications, this procedure is generic. **Not** all illustrations within this procedure will represent all applications.

The exhaust aftertreatment is composed of four sections. These sections are:

1. Aftertreatment DOC
2. Aftertreatment diesel particulate filter (DPF)
3. Decomposition reactor tube
4. Selective catalytic reduction (SCR).

Note : The aftertreatment DOC is integrated into the inlet of the exhaust aftertreatment system.

Note : If the aftertreatment DPF or aftertreatment DOC is being replaced due to progressive damage from engine oil or fuel, clean the downpipe from the turbocharger outlet to the aftertreatment DOC. Refer to Procedure 011-049 in Section 11.

(/qs3/pubsys2/xml/en/procedures/493/493-011-049.html)

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns, or other

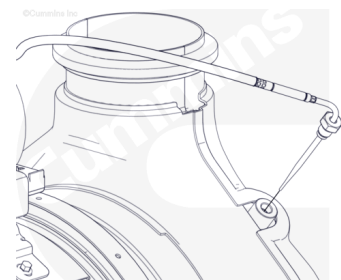
serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the original equipment manufacturer (OEM) heat shield attached to the DOC, if equipped.

Disassemble

⚠ CAUTION ⚠

The aftertreatment DOC intake temperature sensor probe must be removed prior to aftertreatment DOC removal or damage to the sensor wiring and sensor probe will occur.



Remove the aftertreatment DOC intake temperature sensor probe. Refer to Procedure 019-449 in Section 19.

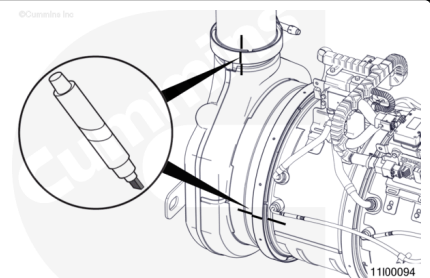
(/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)

Note : The aftertreatment DOC intake temperature sensor probe must be removed prior to aftertreatment DOC removal or damage to the sensor wiring and sensor probe will occur.

Remove

⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use an air tool to remove the V-band clamp nut. An air tool will damage the threads. Apply thread lubricant to the V-band clamp threads prior to nut removal.

Draw an orientation reference line across each of the V-band clamps, aftertreatment canister sections, and connection points to the tailpipe. This will aid in linking up sections and V-band clamps to their original orientation during installation.

Note : If necessary, remove additional mounting hardware to remove the aftertreatment DOC from the vehicle.

Remove the V-band clamps from the inlet and outlet of the aftertreatment DOC.

Remove and discard the gaskets.

Note : Depending upon aftertreatment system installation, the full aftertreatment system may need to be removed from the vehicle prior to disassembly.

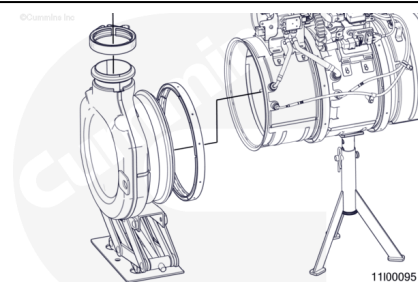
Support the aftertreatment system to prevent it from being freely suspended after the aftertreatment DOC is removed.

Support the aftertreatment DOC catalyst with a lowering jack.

Remove the clamps on either end of the aftertreatment DOC.

Remove the exhaust hangers that support the aftertreatment DOC and slowly remove the canister from the vehicle.

Discard the gasket from the V-band joint(s).



Clean and Inspect for Reuse

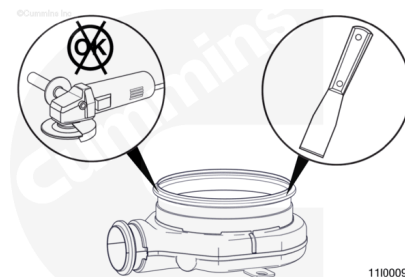
⚠ CAUTION ⚠

Do not use a grinder or abrasive air tool to remove residual gasket material, as this can damage the flange and cause the connection to leak.

⚠ CAUTION ⚠

Do not use an open flame to burn off soot accumulation from the face of the aftertreatment DPF.

⚠ CAUTION ⚠



Do not scrape off soot accumulation from the face of the aftertreatment DPF.

⚠ CAUTION ⚠

Do not steam clean the inlet of the DOC as damage to the aftertreatment DOC can result.

Remove any residual gasket material from the flanges on the aftertreatment DPF, DOC, and catalyst inlet with a putty knife.

Note : Avoid dropping fragments of gasket material into the aftertreatment diesel particulate filter and catalyst.

Inspect the exhaust gas temperature sensor boss threads for damage.

If thread damage is found on the exhaust gas temperature sensor boss threads, a thread repair insert **must** be used for repair.

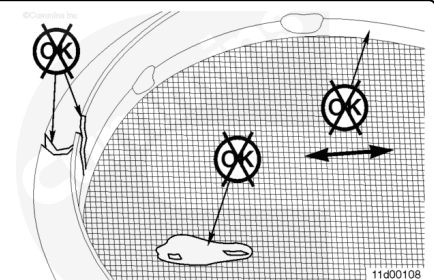
Note : If the aftertreatment diesel particulate filter is being inspected due to progressive damage that introduced engine oil or excessive fuel into the exhaust, inspect the downpipe from the turbocharger outlet to the aftertreatment DOC.

If a trail of engine oil or fuel can be seen exiting the turbocharger outlet, steam clean the downpipe between the turbocharger and the aftertreatment DOC.

If the aftertreatment DOC section has shifted, moved, or is loose inside the canister, replace the aftertreatment DOC section.

Refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 (</qs3/pubsys2/xml/en/bulletin/4021600.html>), to inspect the inlet and outlet faces of the aftertreatment DOC for signs of damage.

Replace the aftertreatment DOC, if it is damaged.



⚠ WARNING ⚠

The material captured in a partial flow diesel particulate filter and/or a diesel particulate filter may contain elevated concentrations of metals. Primarily zinc, molybdenum, and possibly polynuclear aromatic hydrocarbons, that may be regulated. These materials must be characterized, handled, and disposed of according to applicable local regulations. In addition, due to the presence of the above-listed chemicals and other potentially toxic components such as oxides of calcium, zinc, phosphorous, silicon, sulfur, and iron, exhaust filter maintenance must be completed only by appropriately trained personnel.



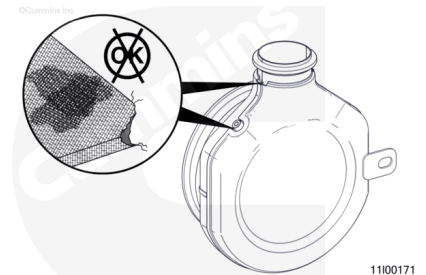
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

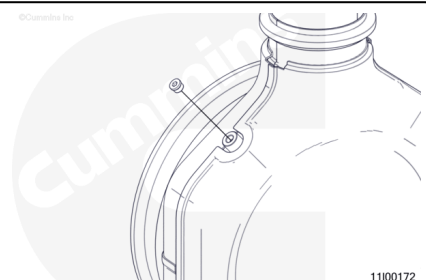
Inspect the outlet of the aftertreatment DOC for loose debris and soot. Use a rag to wipe off any soot. Use a HEPA vacuum to remove any remaining loose debris.

Note : Use a vacuum bag, such as a drywall dust bag, to capture the soot that is removed.

Inspect the aftertreatment DOC insulation for damage.

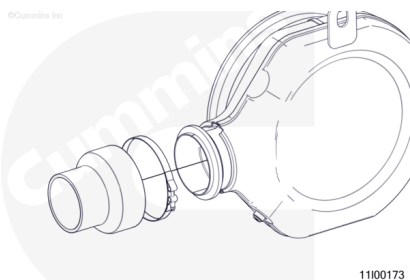


Install a threaded plastic plug, Part Number 4919495, into the exhaust gas temperature sensor port on the DOC. This will prevent air from escaping the DOC.



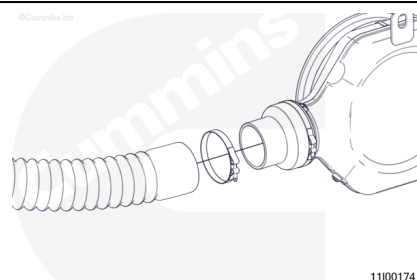
Obtain a 51 to 102 mm [2 to 4 in] rubber pipe reducer.

Install the larger end of the reducer to the inlet of the DOC.



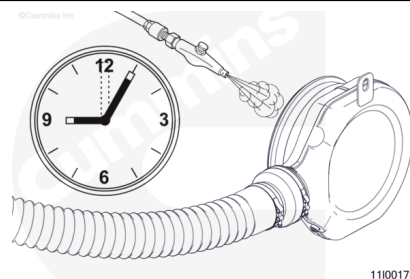
Attach the vacuum hose to the 51 mm [2 in] end of the pipe reducer.

Use tape to seal any open ports.



⚠ CAUTION ⚠

**Use only a rubber-tipped air gun near the catalyst material.
Damage will result if a metal air gun strikes the catalyst material.**



Adjust the air supply for a rubber-tipped air gun to 621 kPa [90 psi].

Switch on the vacuum.

Blow compressed air across the outlet face of the aftertreatment DOC for approximately 5 minutes.

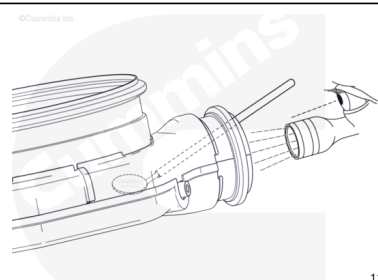
Maintain a 13 to 25 mm [1/2 to 1 in] distance between the air gun and the DOC face. Use a sweeping motion across the entire face of the DOC and attempt to blow air through every cell.

Allow the vacuum to operate for 1 minute after blowing out the aftertreatment DOC.

Switch off the vacuum and inspect the cells of the DOC.

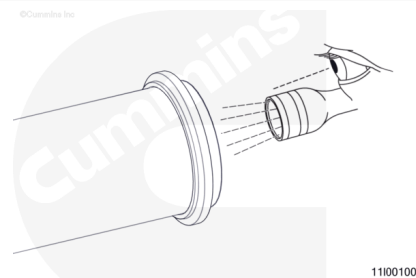
For side-in configurations:

Place an inspection mirror into the inlet of the DOC and shine a flashlight on the mirror to inspect the cells of the inlet face.



For end-in configurations:

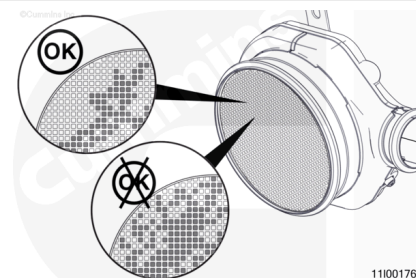
Use a flashlight to inspect the inlet face of the DOC.



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Upon inspection, if less than 50 percent of the cells are blocked (1) the aftertreatment DOC can be reused. If more than 50 percent of the cells remain blocked (2), repeat the cleaning procedure.

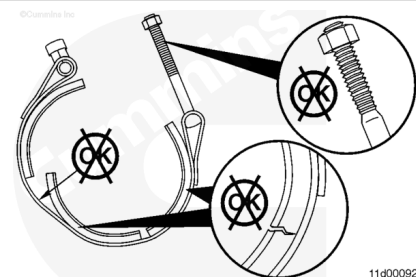
If 50 percent of the cells remain blocked after the second cleaning, replace the DOC.



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Inspect the V-band clamps and aftertreatment DOC mounting straps for signs of overextension. The V-band **must not** be bent or otherwise damaged.

Replace the V-band clamps or aftertreatment DOC mounting straps if damage is found.

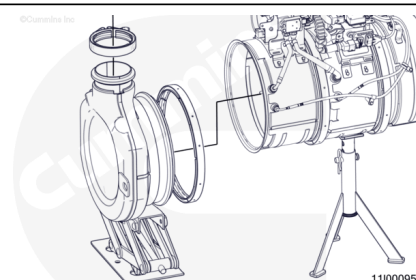


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Install

⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ CAUTION ⚠

Do not use an air tool to tighten the V-band clamp. An air tool will damage the threads.

Service locations in California. See Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787 (</qs3/pubsys2/xml/en/bulletin/6642787.html>).

Install new gaskets on the inlet and outlet of the aftertreatment DOC.

Apply a coat of anti-seize compound, Part Number 3824879, to the threads of the V-band clamps.

Install the aftertreatment DOC.

Tighten the V-band clamp used to secure the aftertreatment DOC to the aftertreatment diesel particulate filter.

Torque Value: 20 n•m [177 in-lb]

Tighten the V-band clamp used to secure the inlet of the aftertreatment DOC to the exhaust pipe.

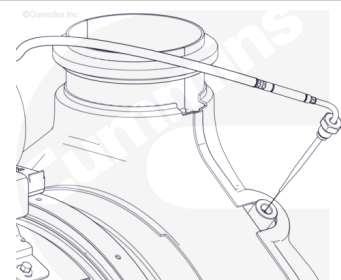
Note : Do not place the T-bolt section of the V-band clamp near the wiring harness.

Torque Value: 13.5 n•m [120 in-lb]

If necessary, install any additional mounting hardware that was used to secure the aftertreatment DOC to the vehicle.

Assemble

Install the exhaust gas temperature sensor probes. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-449.html>)



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Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the OEM wiring harness to the aftertreatment interface harness. Refer to Procedure 019-477 in Section 19. (</qs3/pubsys2/xml/en/procedures/493/493-019-477.html>)
- Install the OEM heat shield to the DOC, if removed.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for fault codes and exhaust leaks.

Last Modified: 26-Aug-2025
